

Problem Definition

SmartSociety — Web-Based Housing Society Management System

Prepared for Contest-AZM / Aptech Learning, Full-Stack Application Development

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1 Background

Housing societies — multi-tower residential complexes with a resident base, a management committee, and a security workforce — run on paper registers, WhatsApp groups, and a security-desk logbook. Maintenance dues are tracked in spreadsheets that fall out of date the moment a payment is collected in cash. Visitor entries are handwritten at the gate, which means there is no reliable record of who is on the premises at any given moment, and no way for a resident to pre-clear a guest without physically calling down to security.

This produces three recurring failure modes: financial dues go uncollected because nobody is tracking who owes what past a given due date; security incidents (an overstaying delivery person, an unverified visitor) go unnoticed because nothing is watching for them between guard shifts; and residents lose trust in the management committee because there is no visible, timestamped record of what was decided, billed, or resolved.

2 Problem Statement

STATEMENT

A housing society with residents, a management committee, and a gate-security team has no single system of record for who lives where, who owes what, who is on the premises, and who is accountable for resolving a complaint — which means every one of those questions currently has to be answered by asking a person, rather than checking a record.

Three roles are affected differently by this gap:

- **Residents** cannot see a current bill breakdown, cannot pre-approve a guest without a phone call, and have no way to track whether a maintenance complaint they raised is actually being worked on.
- **Security guards** have no fast way to verify that a person at the gate was actually pre-approved, and no system that flags a visitor who has overstayed without a guard remembering to check a clock.
- **Society administrators** have no dashboard for occupancy, collections, or open tickets, and no audit trail proving that a given financial adjustment or complaint resolution actually happened the way it was recorded.

3 Objectives

The system set out to replace the informal, per-person process above with a single role-aware application that:

1. Gives every resident a real-time view of their dues, visitor passes, complaints, and amenity bookings, with self-service actions for each.
2. Gives every guard a fast, verifiable way to admit a visitor — by scanned QR code or a manually entered gate-pass code — and surfaces overstay situations automatically rather than relying on a guard noticing.
3. Gives every administrator a single dashboard for occupancy, billing collection, complaint SLAs, and staffing, with the ability to onboard/offboard residents and staff without touching the database directly.
4. Keeps an immutable record of the three categories of activity most likely to be disputed later — gate entries, complaint status changes, and financial edits — so that "what actually happened" is never just one person's word against another's.
5. Broadcasts a genuine emergency (siren) to every connected user in real time, not just to whoever happens to be looking at a noticeboard.

4 Scope

IN SCOPE

Resident, Admin, and Guard portals; visitor gate-pass lifecycle; maintenance billing and penalties; complaint ticketing with SLA and routing; amenity booking; notices and polls; role-based auth with email OTP; immutable audit logging; real-time emergency broadcast; a directory of on-site service staff (plumber, electrician, etc.).

OUT OF SCOPE

Real payment-gateway settlement and bank reconciliation (simulated per the project brief); native mobile apps (the web app is responsive instead); hardware integration with physical turnstiles or biometric scanners; multi-society / multi-tenant deployment.

5 Stakeholders

ROLE	PRIMARY CONCERN	SYSTEM ANSWER
Resident / Flat Owner	Pay dues, admit guests, get complaints resolved	Resident portal
Tenant	Same as resident, without ownership records	Resident portal (occupancy type: Tenant)
Security Guard	Verify who's allowed in, log who's inside	Guard portal
Society Administrator	Collections, staffing, oversight, emergencies	Admin portal
Service staff (plumber, electrician, ...)	Be reachable when needed	Directory entry, no login